



BCD to 7-Segment Display Decoder Circuit

Name: Mariam Ahmed Maher

ID : 202400733

Course: Digital Logic & Computer Architecture

Instructor: Dr. Mahdy Maree

Date: 2024 Fall

Components Used

Component	Quantity
4-Pin DIP Switch	1
Common Anode 7-Segment Display	1
7447 Decoder IC	1
1KΩ Resistors	4
Breadboard	1
Jumper Wires	21

Component Functions

7-Segment Display	Used to display decimal digits from 0 to 9.
7447 Decoder IC	Converts Binary-Coded Decimal (BCD) inputs into signals that drive the 7-segment display.
Resistors	Used for: • Current limiting • Voltage division
4-Pin DIP Switch	Used to provide the 4-bit binary input to the circuit.
Breadboard	Used for circuit prototyping and testing.
Jumper Wires	Used to: • Connect components on the breadboard • Connect the circuit to power and ground

Circuit Connections

DIP Switch Connections	• Pins 1–4 → Power (VCC) • Pins 5–8 → Ground through 1KΩ resistors
7447 IC Connections	• Pin 16 → Power (VCC) • Pin 8 → Ground (GND) • Pins 3, 4, and 5 → Power (VCC)
7-Segment Display Connections	• Pins 3 and 8 → VCC
Switch to 7447 Connections	• Switch Pin 5 → IC Pin 7 • Switch Pin 6 → IC Pin 1 • Switch Pin 7 → IC Pin 2 • Switch Pin 8 → IC Pin 6
7447 to 7-Segment Connections	• IC Pin 15 → Display Pin 9 • IC Pin 14 → Display Pin 10 • IC Pin 13 → Display Pin 7 • IC Pin 12 → Display Pin 6 • IC Pin 11 → Display Pin 4 • IC Pin 10 → Display Pin 2 • IC Pin 9 → Display Pin 1

Final Step

Connect all power rails together and all ground rails together.

Project Outcome

The circuit successfully converts a 4-bit BCD input into its corresponding decimal digit on a common anode 7-segment display using the 7447 decoder IC.